

Gongbo Zhang

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PROFESSIONAL SUMMARY

- **Seven years of NLP and AI experience across academia and industry**, from engineering and research roles at **Google** and **Facebook** to biomedical informatics research at Columbia, directed toward making clinical evidence more accessible, reliable, and usable.
- Led first-author research across the clinical NLP pipeline: **entity recognition** (PICOX, *JAMIA* 2024; FinePICO, *JAMIA* 2025), **concept normalization** (neural-symbolic AI agent, *npj Digit Med* 2026), **evidence summarization** (fine-tuned open-source LLMs, *npj Digit Med* 2024), and **question answering** (BriefContext, *npj Digit Med* 2025; *ACL* 2026).
- Active in professional service: incoming **Editorial Board Member of *JBIM*** (2027); Associate Editor/Advisory Committee member for special issues in *JBIM* and *ACI*; reviewer for 15+ conferences and journals.
- Recipient of the **NIH/NLM K99/R00 Pathway to Independence Award** (2026).

EDUCATION

University of Delaware Ph.D., Computer Science Advisor: Prof. Hagit Shatkay GPA: 3.97	Newark, DE Sep 2015 – Dec 2021
Georgia Institute of Technology M.S., Computer Science	Atlanta, GA Sep 2012 – Dec 2013
Dalian University of Technology B.Eng., Software Engineering	Dalian, China Sep 2008 – Jun 2012

ACADEMIC APPOINTMENTS

Columbia University Irving Medical Center Department of Biomedical Informatics (Weng Lab) Associate Research Scientist	New York, NY Nov 2025 – Present
• Developed a neural-symbolic AI agent pipeline for biomedical concept normalization (<i>npj Digital Medicine</i>, 2026). • Designed a self-corrective agentic mechanism for question answering (<i>ACL</i> 2026, Main Conference).	
Postdoctoral Research Scientist <i>Advisor: Prof. Chunhua Weng</i>	May 2023 – Nov 2025
• Developed named-entity recognition models to extract PICO elements from clinical trials. • Optimized and deployed LLMs for clinical evidence summarization.	

INDUSTRY EXPERIENCE

Google LLC, Google Cloud Software Engineer	Sunnyvale, CA Nov 2020 – Mar 2023
• Developed containerized packages for TPU function verification. • Developed dashboards for monitoring the health state of TPUs in the fleet. • Designed automatic pipelines for checksum generation of TPU health checkers.	
Facebook Inc., Search Team Research Scientist Intern	Bellevue, WA Jun 2019 – Aug 2019
• Developed a query intent model to predict the type of pages users would interact with based on their queries. • Optimized the model via data cleaning strategies and embedding representations of the queries.	
Google LLC, Natural Language Understanding Team Software Engineer Intern	Mountain View, CA Jun 2018 – Aug 2018

- Developed a **neural quality-estimation (QE) network** to automatically estimate post-edit effort for translations.
- Processed training data and applied transfer learning to train the neural model.
- Demonstrated that the model reliably evaluated translations on multiple data sets.

Google LLC, Search Infrastructure Team

Software Engineer Intern

Mountain View, CA

Jun 2017 – Aug 2017

- Participated in the design of a new rendering backend to simplify dependencies of web search features.
- Modularized the rendering backend in Google Web Server to accelerate search feature development.

HONORS & AWARDS

NIH/NLM K99/R00 Pathway to Independence Award	2026
Udel College of Engineering Teaching Assistant Award	2021
Udel Distinguished Graduate Student Award	2020
Udel College of Engineering Fellowship	2020
Udel Doctoral Fellowship Nomination (top 2 candidates, CIS Dept.)	2020
Udel Professional Development Award	2018
Laird Fellowship Nomination (top 10 first-year graduate students, College of Engineering)	2016

PUBLICATIONS

[†]Equal contribution; [‡]Co-corresponding authors.

Journal Articles

- [J1] **A neural-symbolic AI agent system for biomedical concept mapping**
Zhang G., Fang Y., Chen F., Ta C., Hripcsak G., Ryan P., Peng Y., Weng C.
npj Digital Medicine, 2026 Apr; 9(1). doi: 10.1038/s41746-026-02594-6. PMID: 41935204.
- [J2] **A data-driven method for research trend analysis in a scientific discipline: Application to the Journal of Biomedical Informatics**
 Fang Y.[†], **Zhang G.**[†], Tejada S.S., Chen F., Shortliffe E., Patel V.L., Peleg M., Weng C.
Journal of Biomedical Informatics, 2026 Jun; 178:105013. doi: 10.1016/j.jbi.2026.105013. PMID: 41839246.
- [J3] **Evaluating unsupervised and rule-based phenotyping methods versus administrative code counts for systemic sclerosis identification**
 Luo Y., **Zhang G.**, Agarwal A., Weng C., Bernstein E.J.
Seminars in Arthritis and Rheumatism, 2026 Oct; 80:153033. doi: 10.1016/j.semarthrit.2026.153033. PMID: 42385561.
- [J4] **Tutorial: Guidance on the use of large language models for medical research**
 Jin Q., Wan N., Leaman R., Tian S., Wang Z., Yang Y., Wang Z., Xiong G., Lai P.T., Zhu Q., Hou B., Sarfo-Gyamfi M., **Zhang G.**, Gilson A., Bhasuran B., He Z., Zhang A., Sun J., Weng C., Summers R.M., Chen Q., Peng Y., Lu Z.
Nature Protocols, 2026 Jul. doi: 10.1038/s41596-026-01408-z. PMID: 42498804.
- [J5] **A critical evaluation of generative query expansion on biomedical literature retrieval**
 Fang Y., **Zhang G.**, Chen F., Peng Y., Weng C.
Journal of the American Medical Informatics Association, 2026 Jun; 33(6):1121–1133. doi: 10.1093/jamia/ocag037. PMID: 41921511.
- [J6] **Interpretable fine-tuned large language models facilitate making genetic test decisions for rare diseases**
 Nguyen Q.M., Chen F., Liu C., Campbell I.M., **Zhang G.**, Wu D., Szigety K.M., Sheppard S.E., Ahimaz P., Ta C.N., Chung W.K., Weng C., Wang K.
npj Digital Medicine, 2026 May; 9(1). doi: 10.1038/s41746-026-02733-z. PMID: 42156861.

- [J7] **CPGPrompt: Translating clinical guidelines into large language model-executable decision support**
Deng R., Martin G., Wang T., Zhang G., Liu Y., Weng C., Wang Y., Rousseau J.F., Peng Y.
Journal of the American Medical Informatics Association, 2026 Apr; 33(4):855–862.
 doi: 10.1093/jamia/ocag026. PMID: 41746783.
- [J8] **Leveraging long context in retrieval augmented language models for medical question answering**
Zhang G., Xu Z., Jin Q., Chen F., Fang Y., Liu Y., Rousseau J.F., Xu Z., Lu Z., Weng C., Peng Y.
npj Digital Medicine, 2025 May; 8(1):239. doi: 10.1038/s41746-025-01651-w. PMID: 40316710.
- [J9] **Scalable scientific interest profiling using large language models**
Liang Y.[†], Zhang G.[†], Sun E., Idnay B., Fang Y., Chen F., Ta C., Peng Y.[‡], Weng C.[‡]
Journal of Biomedical Informatics, 2025 Dec; 172:104949. doi: 10.1016/j.jbi.2025.104949. PMID: 41177243.
- [J10] **Semi-supervised learning from small annotated data and large unlabeled data for fine-grained Participants, Intervention, Comparison, and Outcomes entity recognition**
Chen F.[†], Zhang G.[†], Fang Y., Peng Y.[‡], Weng C.[‡]
Journal of the American Medical Informatics Association, 2025 Mar; 32(3):555–565.
 doi: 10.1093/jamia/ocae326. PMID: 39823371.
- [J11] **Mini-mental status examination phenotyping for Alzheimer’s disease patients using both structured and narrative electronic health record features**
Idnay B.[†], Zhang G.[†], Chen F., Ta C.N., Schelke M.W., Marder K., Weng C.
Journal of the American Medical Informatics Association, 2025 Jan; 32(1):119–128.
 doi: 10.1093/jamia/ocae274. PMID: 39520712.
- [J12] **CLEAR: A vision to support clinical evidence lifecycle with continuous learning**
Fang Y., Zhang G., Chen F., Hripcsak G., Peng Y., Ryan P., Weng C.
Journal of Biomedical Informatics, 2025 Sep; 169:104884. doi: 10.1016/j.jbi.2025.104884. PMID: 40744316.
- [J13] **Closing the gap between open-source and commercial large language models for medical evidence summarization**
Zhang G., Jin Q., Zhou Y., Wang S., Idnay B., Luo Y., Park E., Nestor J.G., Spotnitz M.E., Soroush A., Campion T.R. Jr, Lu Z., Weng C.[‡], Peng Y.[‡]
npj Digital Medicine, 2024 Sep; 7(1):239. doi: 10.1038/s41746-024-01239-w. PMID: 39251804.
- [J14] **Leveraging generative AI for clinical evidence synthesis needs to ensure trustworthiness**
Zhang G., Jin Q., McInerney D.J., Chen Y., Wang F., Cole C.L., Yang Q., Wang Y., Malin B.A., Peleg M., Wallace B.C., Lu Z., Weng C.[‡], Peng Y.[‡]
Journal of Biomedical Informatics, 2024 May; 153:104640. doi: 10.1016/j.jbi.2024.104640. PMID: 38608915.
- [J15] **A span-based model for extracting overlapping PICO entities from randomized controlled trial publications**
Zhang G., Zhou Y., Hu Y., Xu H., Weng C.[‡], Peng Y.[‡]
Journal of the American Medical Informatics Association, 2024 Apr; 31(5):1163–1171.
 doi: 10.1093/jamia/ocae065. PMID: 38471120.
- [J16] **Rare disease diagnosis using knowledge-guided retrieval augmentation for ChatGPT**
Zelin C., Chung W.K., Jeanne M., Zhang G., Weng C.
Journal of Biomedical Informatics, 2024 Sep; 157:104702. doi: 10.1016/j.jbi.2024.104702. PMID: 39084480.

- [J17] **Criteria2Query 3.0: Leveraging generative large language models for clinical trial eligibility query generation**
Park J.[†], Fang Y.[†], Ta C., Zhang G., Idnay B., Chen F., Feng D., Shyu R., Gordon E.R., Spotnitz M., Weng C.
Journal of Biomedical Informatics, 2024 Jun; 154:104649. doi: 10.1016/j.jbi.2024.104649. PMID: 38697494.
- [J18] **A survey of clinicians' views of the utility of large language models**
Spotnitz M., Idnay B., Gordon E.R., Shyu R., Zhang G., Liu C., Cimino J.J., Weng C.
Applied Clinical Informatics, 2024 Mar; 15(2):306–312. doi: 10.1055/a-2281-7092. PMID: 38442909.
- [J19] **Translational drug-interaction corpus**
Zhang S., Wu H., Wang L., Zhang G., Rocha L.M., Shatkay H., Li L.
Database (Oxford), 2022 May; 2022:baac031. doi: 10.1093/database/baac031. PMID: 35616099.
- [J20] **Utilizing image and caption information for biomedical document classification**
Li P., Jiang X., Zhang G., Trabucco J.T., Raciti D., Smith C., Ringwald M., Marai G.E., Arighi C., Shatkay H.
Bioinformatics (ISMB Proceedings), 2021 Jul; 37(Suppl_1):i468–i476.
doi: 10.1093/bioinformatics/btab331. PMID: 34252939.
- [J21] **An effective biomedical document classification scheme in support of biocuration: Addressing class imbalance**
Jiang X., Ringwald M., Blake J.A., Arighi C., Zhang G., Shatkay H.
Database (Oxford), 2019 Jan; 2019:baz045. doi: 10.1093/database/baz045. PMID: 31032839.
- [J22] **On the benefits of digital libraries of case studies of analogical design: Documentation, access, analysis, and learning**
Goel A., Zhang G., Wiltgen B., Zhang Y., Vattam S., Yen J.
AI EDAM (Artificial Intelligence for Engineering Design, Analysis and Manufacturing), 2015 May; 29(2):215–227.

Conference Proceedings

- [C1] **Improving retrieval-augmented generation without taxonomy-based error categorization**
Zhang G., Peng Y., Weng C.
Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL), 2026 Jul; pp. 155–165. Main Conference. PMID: 42382679.
- [C2] **EvidenceOutcomes: A dataset of clinical trial publications with clinically meaningful outcomes**
Zhou Y., Newbury A.M., Zhang G., Idnay B.R., Liu H., Weng C., Peng Y.
Studies in Health Technology and Informatics (MEDINFO 2025), 2025 Aug; 329:723–727.
doi: 10.3233/SHTI250935. PMID: 40775953.
- [C3] **Natural language processing in support of evidence-based medicine: A scoping review**
Xu Z., Ma H., Zhang G., Ding Y., Weng C., Peng Y.
Findings of the Association for Computational Linguistics: ACL 2025, 2025 Jul; pp. 21421–21443.
doi: 10.18653/v1/2025.findings-acl.1103.
- [C4] **Clinical note structural knowledge improves word sense disambiguation**
Chen F.[†], Zhang G.[†], Chen S., Callahan T., Weng C.
AMIA Joint Summits on Translational Science Proceedings, 2024 May; pp. 515–524. PMID: 38827062.

- [C5] **Identifying experimental evidence in biomedical abstracts relevant to drug-drug interactions**
Zhang G., Roychowdhury D., Li P., Wu H.Y., Zhang S., Li L., Shatkay H.
Proceedings of the ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB), 2018 Aug; pp. 414–418.
- [C6] **Identifying articles relevant to drug-drug interaction: Addressing class imbalance**
Zhang G., Bhattacharya M., Wu H.Y., Li P., Li L., Shatkay H.
Proceedings of the IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2017 Nov; pp. 1141–1147.
- [C7] **Closing the gap: Domain adaptation from explicit to implicit discourse relations**
 Ji Y., **Zhang G.**, Eisenstein J.
Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), 2015 Sep; pp. 2219–2224.
- [C8] **The Design Study Library: Compiling, analyzing and using biologically inspired design case studies**
 Goel A., **Zhang G.**, Wiltgen B., Zhang Y., Vattam S., Yen J.
Proceedings of Design Computing and Cognition (DCC'14), Springer, 2015; pp. 625–643.

INVITED TALKS & TUTORIALS

Towards a Human-AI Collaborative Medical Research Life Cycle Tutorial, AMIA Annual Symposium	Nov 2025
LLM Applications in Biomedical Text Mining Guest Lecture, G4003: Symbolic AI in Healthcare, Columbia University	Sep 2025
Generative AI Plays a Role in Modern Evidence-Based Medicine Keynote, IEEE/ACM CHASE Conference	Jun 2025
Modern Literature Evidence Computing with LLMs Biologics Effectiveness and Safety (BEST) Initiative Meeting, U.S. FDA	Mar 2025
Neural-Symbolic Methods for Clinical Evidence Guest Lecture, G4003: Symbolic AI in Healthcare, Columbia University	Oct 2024
Clinical Evidence Extraction and Summarization Clinical & Translational Science Center Seminar, Weill Cornell Medicine	Apr 2024
LLMs for Modern Clinical Evidence Summarization BENGAL Proposers' Day, IARPA	Oct 2023

TEACHING

University of Delaware, Dept. of Computer and Information Sciences Graduate Teaching Assistant	Newark, DE
CISC 436/636: Computational Biology and Bioinformatics	Fall 2020
CISC 320: Introduction to Algorithms	Fall 2020
CISC 220: Data Structures	Spring 2016
CISC 106: General Computer Science for Engineers	Fall 2015

PROFESSIONAL SERVICE

Editorial Roles

- Editorial Board Member, *Journal of Biomedical Informatics* Starting Jan 2027
- Associate Editor, *Applied Clinical Informatics* Special Topic: “Artificial Intelligence Application in Clinical Decision Support Systems”
- Advisory Committee, *Journal of Biomedical Informatics* Special Issue: “Agentic AI for Lifecycle-Aware Real-World Evidence: Methods for Scalable, Trustworthy Evaluation of Treatment Effectiveness and Safety”
- Early Career Editorial Board, *Journal of Biomedical Informatics* Jan 2025 – Dec 2026
- Early Career Editorial Board, *Applied Clinical Informatics* Jan 2025 – Dec 2026

Conference Reviewer

- Annual Meeting of the Association for Computational Linguistics (ACL)

- Conference of the European Chapter of the Association for Computational Linguistics (EACL)
- Conference on Empirical Methods in Natural Language Processing (EMNLP)
- Advances in Neural Information Processing Systems (NeurIPS)
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

Journal Reviewer

- *Nature Communications*
- *Scientific Reports*
- *npj Digital Medicine*
- *npj Health Systems*
- *Journal of the American Medical Informatics Association (JAMIA)*
- *Journal of Biomedical Informatics (JBI)*
- *Applied Clinical Informatics (ACI)*
- *Journal of Medical Internet Research (JMIR)*
- *JMIR Medical Informatics*
- *IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)*
- *BMC Medical Informatics and Decision Making*
- *BMC Public Health*
- *BMC Pregnancy and Childbirth*

TECHNICAL SKILLS

LLM post-training: SFT, LoRA/QLoRA, RLHF, DPO, GRPO

Agentic & retrieval systems: neural-symbolic agent systems; retrieval-augmented generation (RAG) and agentic RAG; BM25 and dense retrieval (bi-encoders, cross-encoder reranking); multi-stage literature screening

Inference & serving: vLLM for large-scale inference; Ollama for local model serving; PyTorch, Hugging Face Transformers

Evaluation: rule-based checkers for NLP pipelines; LLM judges; held-out benchmark construction

Programming: Python, SQL, C/C++, Java

Languages: Chinese (native), English (bilingual proficiency)